

DeepSeek-V2

A Strong, Economical, and Efficient MoE Language Model

236B Total Parameters | 21B Activated | 128K Context

arXiv: 2405.04434

The LLM Scaling Dilemma

- Larger models yield better intelligence but incur prohibitive training costs
- KV cache bottleneck: full key-value pairs stored per head per token during autoregressive generation
- Alternatives like GQA and MQA reduce cache but sacrifice model performance

Solution needed → MoE for efficient training + novel attention for efficient inference

DeepSeek-V2 Architecture

Multi-Head Latent Attention (MLA)

Replaces standard MHA with low-rank KV compression

93.3%

KV Cache Reduction

DeepSeekMoE

Replaces dense FFN with shared + routed expert layers

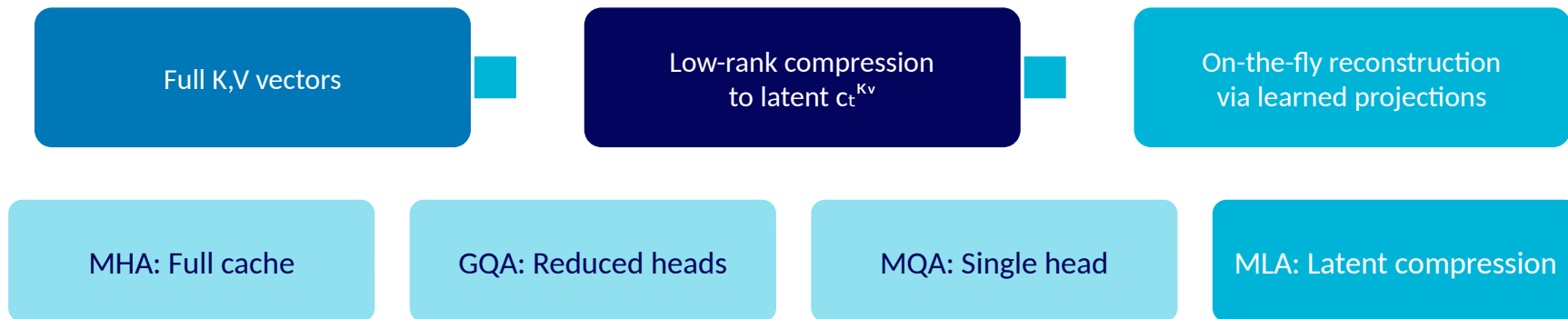
42.5%

Training Cost Reduction

Combined in each Transformer block for maximum efficiency

Multi-Head Latent Attention (MLA)

Compresses KV Cache by 93.3%



Decoupled RoPE enables position-aware reconstruction without caching positional keys

DeepSeekMoE Architecture

Fine-Grained Experts with Shared Expert Isolation

Shared Experts (N_s)

Always activated; capture common cross-domain knowledge

Routed Experts (N_r)

Sparsely activated via top-K, router for specialized processing

- Fine-grained segmentation increases specialization
- Device-limited routing (M devices) reduces cross-node communication
- Auxiliary losses enforce load balance; token-dropping prevents overflow during training

Efficiency: DeepSeek-V2 vs. DeepSeek 67B

42.5%

Training Cost Reduction

236B MoE trained cheaper than
67B dense

93.3%

KV Cache Reduction

MLA latent compression vs.
standard MHA

5.76×

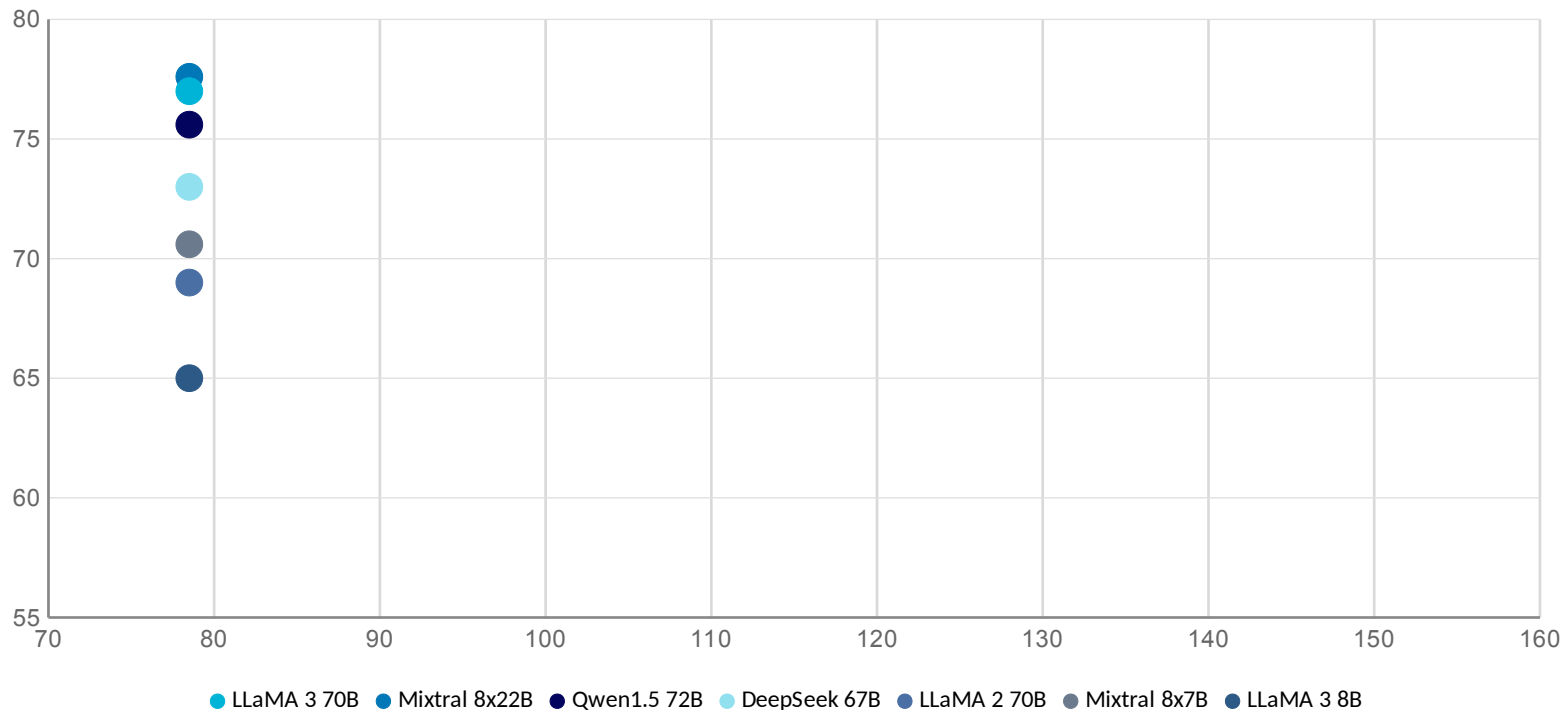
Generation Throughput

Maximum throughput
improvement

DeepSeek-V2 (236B total / 21B activated) achieves superior performance at a fraction of the cost of DeepSeek 67B (dense)

MMLU vs. Activated Parameters

DeepSeek-V2: Pareto-Optimal Efficiency



Alignment: DeepSeek-V2 Chat (RL)

Post-training: SFT on 1.5M conversational sessions + GRPO-based reinforcement learning

AlpacaEval 2.0

38.9%

Length-controlled win rate

MT-Bench

8.97

Overall score

AlignBench

7.91

Overall (Chinese)

Outperforms all open-source and most closed-source models on Chinese benchmarks

Key Takeaways

- 1 MLA achieves superior performance to MHA while reducing KV cache by 93.3%
- 2 DeepSeekMoE enables training 236B params at 42.5% lower cost than 67B dense
- 3 Only 21B activated parameters needed to match or exceed 70B-class dense models
- 4 5.76× throughput improvement makes DeepSeek-V2 practical for deployment
- 5 Strongest open-source chat model in Chinese, competitive in English